

Michał Andrzej Sitarz

(+385) 99 214 3812 | michal@sitarz.org.pl | [michal-sitarz2.github.io](https://github.com/michal-sitarz2)

Education

KTH Royal Institute of Technology , <i>MSc Machine Learning</i> , Sweden	Aug 2023 - Mar 2026
<i>Relevant Modules:</i> Machine Learning, Deep Learning, Computer Vision, Robotics, Graphics	
University of Surrey , <i>BSc (Hons) Computer Science</i> , UK	Aug 2019 - Jun 2023
<i>Relevant Modules:</i> Deep Learning, Artificial Intelligence, Software Engineering, NLP	

Experience

National Institute of Informatics , <i>Research Intern</i> , Japan	Aug 2025 - Jan 2026
- Analyzed the reflectance behavior of synthetic materials using BRDF parameters and Video Motion Magnification techniques, producing promising results that will inform future lab research. - Developed Python implementations of signal processing algorithms, including Riesz Transform for motion/reflectance decomposition.	
Ericsson Research , <i>MSc Thesis Student</i> , Sweden	Jan 2025 - Jul 2025
- Applied adaptive RL with quantization to network-aware robot control, demonstrating improved model efficiency and policy interpretability on proprietary datasets. - Extended the base algorithm with stability improvements and alignment with real-world control constraints. - Built end-to-end training and evaluation pipelines for network-constrained environments.	
KTH Royal Institute of Technology , <i>Teaching Assistant in Visualization</i> , Sweden	Aug 2024 - Nov 2024
Graded coursework and led support sessions, guiding students through core visualization concepts.	
University of Surrey (CVSSP Lab) , <i>Summer Research Intern</i> , UK	Jul 2022 - Sep 2022
- Built a Gazebo simulation environment and a curriculum-based Reinforcement Learning pipeline using PyTorch. - Trained a TurtleBot agent to autonomously score goals in simulation, supporting future deployment for RoboCup.	
Hawk-Eye Innovations , <i>Machine Learning Engineer (Placement)</i> , UK	Sep 2021 - Jun 2022
- Improved UX and functionality of an internal React-based annotation tool used for training models. - Developed C++ applications for data collection, visualization, and preprocessing used by tennis operators. - Optimized large-scale data-processing pipelines, significantly reducing preparation time for training tennis models. - Prototyped ML pipeline improvements during hackathons, including evaluating YOLO-based pose estimation models in PyTorch to explore potential gains in accuracy and latency.	

Projects

Physics-Based 3D Gaussian Splatting: Coupled 3DGS reconstruction with MPM simulation. Created a custom gsplat-based trainer in PyTorch with densification, ported my C++ MPM solver to NVIDIA Warp.	2026
Generative AI Reproductions: Reproduced and ablated flow-based models (Minibatch Optimal Transport), diffusion-based segmentation, and VAEs, gaining hands-on experience with modern generative AI	2024
Autonomous Navigation: Implemented Hybrid A* path planning with PID control for virtual car navigation in Unity, handling dynamic obstacle avoidance	2024
Skin Cancer Classification: Experimented with CNN architectures (ResNet, EfficientNet, etc.) on the HAM10000 dataset, exploring data augmentation and transfer learning techniques	2023
RoboKinesis (BSc Thesis): Developed a vision-based robotic control system in ROS, combining keypoint tracking with inverse kinematics for gesture-driven manipulation	2023
Genetically Modified Wolf Optimization for Deep Neural Nets: Developed a novel optimization approach combining genetic algorithms and SGD; documented the methodology in a technical paper (publication)	2023

Skills

Programming Languages: Python, C++, Java, React
ML/AI Frameworks: PyTorch, OpenCV, NumPy, scikit-learn, JAX, CUDA, NVIDIA Warp
Tools & Infrastructure: Git, CI/CD, Docker, Linux, TorchServe, SQL, REST APIs
3D/Simulation: ROS, Gazebo, Blender, Unreal Engine 5, Unity, OpenGL, GLSL
Languages: Polish (Native), English (C2), Croatian (C2)